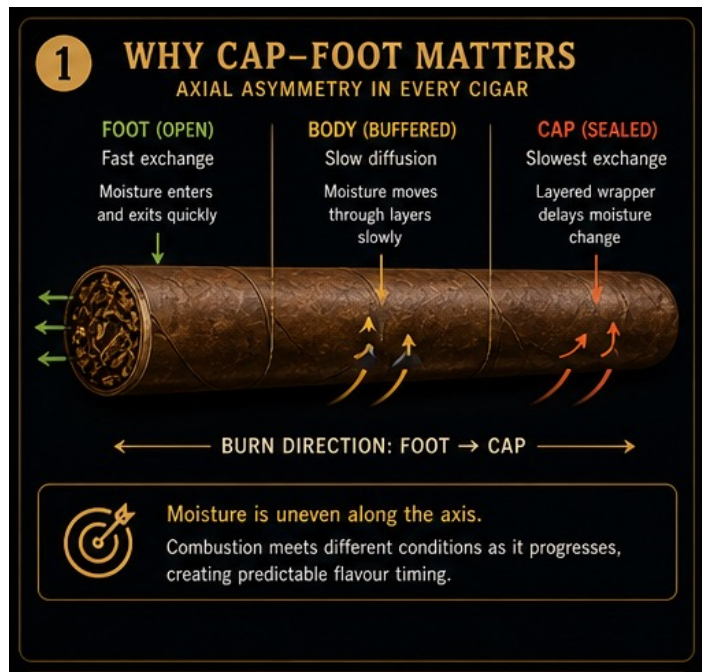


Cap-Foot Diagnostics & Flavour Timing

Module purpose · predictive diagnostics for professional cigar service



Module 5 objective

- Measure and interpret moisture differences between cap and foot.
- Predict flavour timing before lighting.
- Diagnose axial imbalance and recommend corrective actions.
- Manage cigar performance using evidence rather than subjective opinion.

Measure. Interpret. Predict. Correct.

Module 5 Learning Map

What this module covers

FOUNDATIONS

- Axial equalization
- Cap and foot measurement
- Cap-foot delta
- Magnitude and direction
- Under- vs over-humidified core

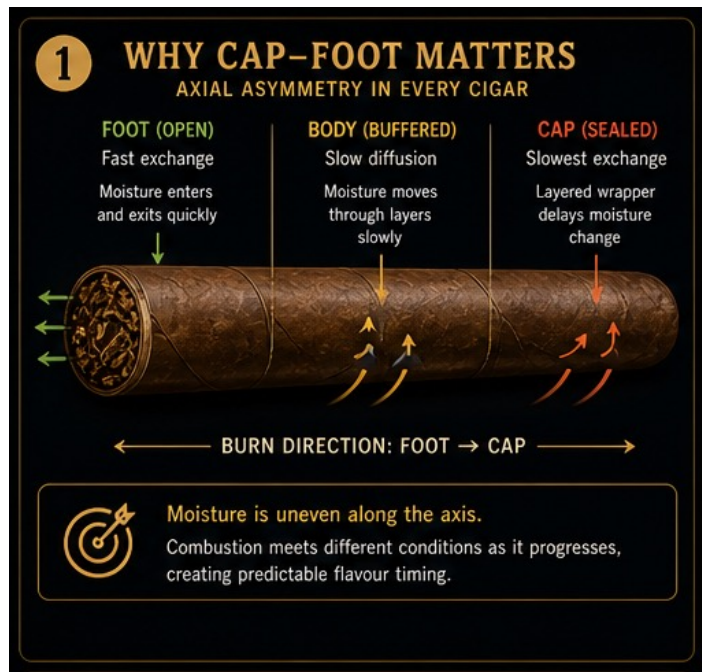
APPLICATION

- Flavour timing profiles
- Density and blender effects
- Vitola and airflow tuning
- Professional scenarios
- Examination alignment

The goal is to predict the smoking journey before ignition.

Why Cap-Foot Matters

A cigar is an asymmetric moisture system



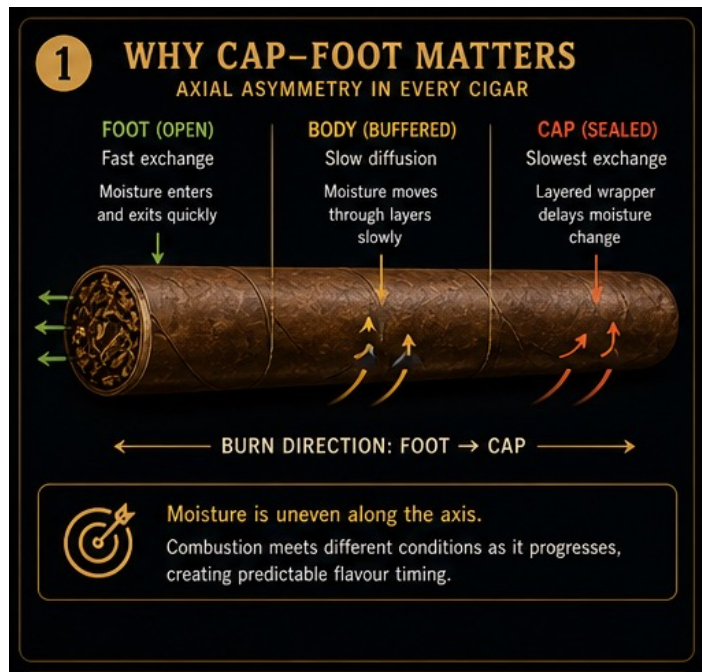
Why this matters professionally

- Foot: open structure and fast moisture exchange.
- Body: insulated; moisture diffuses through layers more slowly.
- Cap: sealed and layered; typically the slowest exchange area.
- Natural gradients create different combustion conditions along the cigar.

Axial moisture imbalance creates predictable flavour timing.

Why Moisture Moves Unevenly

Equalization is directional and slow



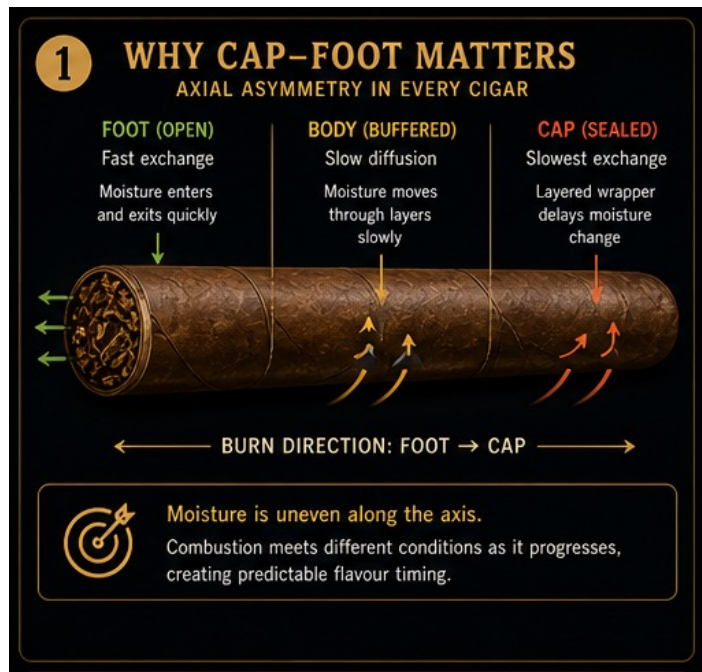
The asymmetric system

- Moisture enters and exits most quickly at the foot.
- The body buffers environmental change.
- The cap/core can lag after RH adjustments.
- Equalization time depends on density, geometry and recent conditioning.

A stable humidor reading does not prove a stable cigar core.

Why This Changes Flavour Timing

The burn travels foot → body → cap



As the ember advances

- Combustion meets different moisture conditions.
- Combustion temperature can change during the smoke.
- Sweetness, strength and aroma can arrive too early or too late.
- The progression can often be anticipated before lighting.

Moisture distribution is part of flavour progression.

How to Measure Properly

Professional standard

2 HOW TO MEASURE PROPERLY

PROFESSIONAL STANDARD

1 **Measure at Foot**
Insert probe 5–8 mm into the foot.

2 **Measure at Cap**
Insert probe 5–8 mm through the cap.

3 **Wait for Stabilization**
Hold until the reading stops changing. (5–10 seconds).

4 **Record**
Note values with timestamp and environment.

COMMON MEASUREMENT ERRORS

Measuring too early
Creates false gradients.

Measuring only the foot
Misses cap (core) information.

Ignoring temperature
Leads to misinterpreted RH readings.

Inconsistent probe depth
Results in inaccurate comparisons.

★ **Consistency = Accuracy = Trust in Your Diagnosis**

Standard procedure

- Use a calibrated leaf-level RH probe.
- Measure the foot and the cap area consistently.
- Insert to a consistent depth, approximately 5–8 mm where appropriate.
- Wait until the reading stabilizes; record RH, time and environment.

Consistency = accuracy = trust in diagnosis.

Measurement Objectives

What the test can and cannot tell you

2 HOW TO MEASURE PROPERLY

PROFESSIONAL STANDARD

- Measure at Foot**
 Insert probe 5–8 mm into the foot.
- Measure at Cap**
 Insert probe 5–8 mm through the cap.
- Wait for Stabilization**
 Hold until the reading stops changing. (5–10 seconds).
- Record**
 Note values with timestamp and environment.

COMMON MEASUREMENT ERRORS

- Measuring too early**
 Creates false gradients.
- Measuring only the foot**
 Misses cap (core) information.
- Ignoring temperature**
 Leads to misinterpreted RH readings.
- Inconsistent probe depth**
 Results in inaccurate comparisons.

★ Consistency = Accuracy = Trust in Your Diagnosis

Cap-foot measurement estimates

- Internal moisture distribution.
- Axial equilibration status.
- Presence and direction of imbalance.
- It complements rather than replaces long-term humidor management.

Use the measurement as a diagnostic snapshot, not a storage substitute.

Common Measurement Errors

Poor technique creates false conclusions

2 HOW TO MEASURE PROPERLY

PROFESSIONAL STANDARD

- Measure at Foot**
 Insert probe 5–8 mm into the foot.
- Measure at Cap**
 Insert probe 5–8 mm through the cap.
- Wait for Stabilization**
 Hold until the reading stops changing. (5–10 seconds).
- Record**
 Note values with timestamp and environment.

COMMON MEASUREMENT ERRORS

- Measuring too early**
 Creates false gradients.
- Measuring only the foot**
 Misses cap (core) information.
- Ignoring temperature**
 Leads to misinterpreted RH readings.
- Inconsistent probe depth**
 Results in inaccurate comparisons.

★ Consistency = Accuracy = Trust in Your Diagnosis

Avoid these errors

- Measuring immediately after removal can create false gradients.
- Measuring only the foot misses core behaviour.
- Ignoring temperature can distort interpretation.
- Inconsistent probe depth makes comparisons unreliable.

Repeatability and controlled conditions matter.

Cap-Foot Delta

The key diagnostic metric

3 CAP-FOOT DELTA THE KEY DIAGNOSTIC METRIC		
DELTA = CAP RH – FOOT RH (Absolute Difference)		
DELTA RANGE	INTERPRETATION	SMOKING PREDICTION
≤ 1.0%	Fully Equalized	Balanced from first light. Optimal experience.
1.0 – 1.5%	Acceptable	Minor flavour timing shift. Small impact.
1.5 – 2.0%	Transitional	Flavour timing shift likely. Experience will change across the smoke.
> 2.0%	Unequalized	Imbalance likely. Correct before smoking.
 Delta is one of the most predictive indicators of cigar readiness and balance.		

Definition

- Delta = | cap RH – foot RH |.
- Magnitude indicates the degree of axial imbalance.
- Direction indicates how the imbalance is likely to behave during smoking.
- Together they predict readiness, flavour timing and combustion balance.

Always read magnitude and direction together.

Interpreting Delta Magnitude

A practical readiness framework

3 CAP-FOOT DELTA THE KEY DIAGNOSTIC METRIC		
DELTA = CAP RH – FOOT RH (Absolute Difference)		
DELTA RANGE	INTERPRETATION	SMOKING PREDICTION
≤ 1.0%	Fully Equalized	Balanced from first light. Optimal experience.
1.0 – 1.5%	Acceptable	Minor flavour timing shift. Small impact.
1.5 – 2.0%	Transitional	Flavour timing shift likely. Experience will change across the smoke.
> 2.0%	Unequalized	Imbalance likely. Correct before smoking.
 Delta is one of the most predictive indicators of cigar readiness and balance.		

Diagnostic ranges

- ≤ 1.0%: fully equalized; balanced from first light.
- 1.0–1.5%: acceptable; minor flavour timing shift.
- 1.5–2.0%: transitional; delayed or uneven flavour likely.
- > 2.0%: unequalized; significant imbalance likely.

Delta is one of the strongest readiness indicators.

Why Direction Matters

The same delta can produce opposite journeys



Two different states

- Foot > Cap: core/cap is relatively drier; late-bloom tendency.
- Cap > Foot: core/cap is relatively wetter; front-loaded tendency.
- Magnitude alone cannot distinguish these behaviours.
- Professional diagnosis must record both readings—not just the difference.

Two scenarios. Two outcomes.

Foot > Cap: Under-Humidified Core

Example: foot 62% · cap 59%



Predicted smoke profile

- First third: brighter, sharper, more wrapper-forward.
- Middle third: strength can rise; sweetness arrives later.
- Final third: heat buildup and bitterness risk increase.
- Smoker may perceive the cigar as stronger and increasingly aggressive.

A dry core can make flavour “open late.”

Foot > Cap: Root Causes

Why cap lag develops



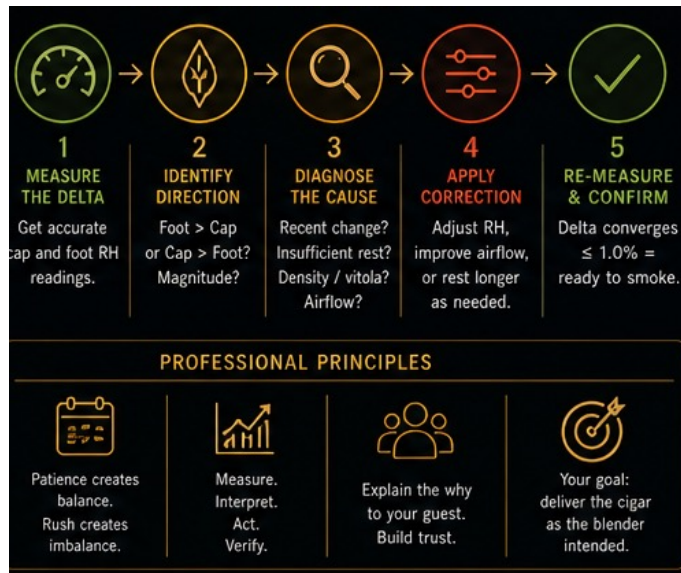
Typical causes

- Recent RH increase.
- Insufficient equilibration time.
- Dense filler resisting moisture diffusion.
- Outer regions respond before the internal core.

Recent conditioning history is part of the diagnosis.

Foot > Cap: Corrective Actions

Restore convergence without overcorrecting



Recommended approach

- Allow additional resting time.
- Maintain stable target RH.
- Avoid aggressive dry-boxing.
- Recheck cap-foot readings after 48–72 hours.

Patience creates balance; rapid correction can create a new gradient.

Cap > Foot: Over-Humidified Core

Example: foot 60% · cap 63%



Predicted smoke profile

- First third: creamy texture, muted flavour, slower ignition.
- Middle third: flavour opens and sweetness improves.
- Final third: muddiness, draw heaviness and declining efficiency may appear.
- Smoker may perceive smoothness without full flavour definition.

Smooth is not automatically equalized.

Cap > Foot: Root Causes

Why the core remains wetter



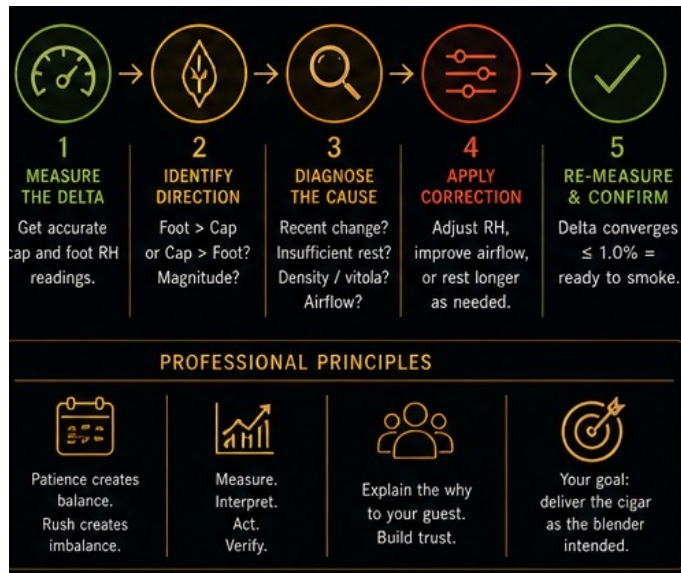
Typical causes

- Recent RH reduction.
- Dry-boxing or lower-RH staging.
- Wrapper/outer layers losing moisture faster than the core.
- The surface can respond before internal moisture catches up.

Direction reveals the recent moisture journey.

Cap > Foot: Corrective Actions

Let the gradient converge



Recommended approach

- Allow short stabilization at lower RH.
- Improve airflow where appropriate.
- Avoid excessive drying.
- Monitor delta convergence before service.

Correct the gradient, not merely the surface reading.

Three Flavour Timing Profiles

Predict the journey before you light



Diagnostic profiles

- Balanced: cap $\approx$ foot; immediate aromatic release and stable progression.
- Late bloom: foot > cap; sharper start, sweetness and expression arrive later.
- Front-loaded: cap > foot; rich early delivery followed by flattening or muddiness.
- Delta turns moisture data into a smoking prediction.

Control the delta. Control the experience.

Balanced Profile

Delta ≤ 1.0%



What to expect

- Balanced ignition.
- Aromatics present immediately.
- Smooth, consistent progression across thirds.
- No major flavour timing shift attributable to axial moisture imbalance.

Fully equalized inventory is the safest choice for immediate service.

Late-Bloom Profile

Foot > cap · delta > 1.0%



The cigar may improve as the ember reaches wetter tobacco.

Front-Loaded Profile

Cap > foot · delta > 1.0%



What to expect

- Rich, smooth and full early delivery.
- Flavour may flatten slightly through the middle.
- Final third can become muddy or combustion can slow.
- The wrapper/outer region has responded faster than the filler core.

Early richness can conceal incomplete equalization.

Density Changes Equalization Speed

Not every cigar converges at the same rate



Density effects

- Dense cigars equalize more slowly and retain moisture longer.
- Dense filler can produce stronger cap-lag effects.
- Delicate cigars respond quickly to RH changes.
- Delicate structures can also have smaller tolerance windows.

Physical construction determines how fast conditioning reaches the core.

Ring Gauge: The Dominant Geometry Variable

Vitola modifies combustion and RH response

NARROW ≤ 46 RG

- Faster airflow and heat buildup.
- Less filler buffering.
- More wrapper-forward and dryness-sensitive.
- Typical adjustment: +0.5% to +1.0% RH.

THICK ≥ 54 RG

- Slower airflow and cooler combustion.
- Greater moisture buffering.
- More filler-dominant combustion.
- Typical adjustment: -0.5% to -1.0% RH.

Peak RH compensates for combustion geometry.

Length & Wrapper Dominance

Secondary geometry still matters

LONGER / THINNER

- Longer cigars accumulate more heat late.
- Typical long-vitola adjustment: +0.25% to +0.5% RH.
- Thin ring gauges increase wrapper influence and aromatic sensitivity.

SHORTER / THICKER

- Short cigars deliver flavour faster with less thermal buildup.
- Thick ring gauges increase filler dominance and buffering.
- The same blend can behave differently across vitolas.

Same blend. Different geometry. Different combustion behaviour.

Draw Resistance & Airflow Tuning

Airflow changes combustion temperature



Practical tuning logic

- If airflow is too fast: combustion overheats and flavour sharpens → RH may need to rise slightly.
- If airflow is too slow: combustion cools and flavour mutes → RH may need to fall slightly.
- Geometry, draw and moisture must be interpreted as one system.
- Avoid using a single RH target mechanically across all vitolas.

Peak RH is a performance setting, not a universal storage number.

Why This Matters for Lounges

One humidor setting does not equal one service condition



Professional lounge practice

- Do not assume every vitola is ready because the cabinet is stable.
- Prepare according to geometry, combustion behaviour and RH response.
- Use cap-foot readings to verify readiness after staging.
- Separate storage conditions from service preparation where necessary.

Professional preparation reduces variability blamed on the cigar.

Scenario 1 – Immediate Flavour Delivery

Customer: “I want the cigar to perform immediately.”



Professional response

- Select inventory with $\leq 1\%$ delta.
- Avoid cigars that were recently adjusted.
- Prioritize fully equalized stock.
- Explain that readiness is verified, not assumed from cabinet RH.

Choose convergence when the guest wants immediate expression.

Scenario 2 – Mid-Smoke Harshness

Customer: “The cigar became harsh halfway through.”



Professional diagnosis

- Consider cap lag or incomplete axial equalization.
- Check whether foot > cap before the next service.
- Explain the moisture mechanism rather than blaming the blend.
- Recommend additional resting time and re-measurement.

The smoking result is evidence of the cigar's moisture history.

Scenario 3 – New Shipment Protocol

Do not serve immediately by default



Professional protocol

- Allow initial stabilization after arrival.
- Measure progressively rather than relying on ambient RH.
- Avoid immediate service while large deltas remain.
- Wait until cap-foot values converge to the desired readiness threshold.

New inventory needs stabilization before evaluation.

Examination Alignment

What candidates must be able to do

DELTA = CAP RH – FOOT RH (Absolute Difference)		
DELTA RANGE	INTERPRETATION	SMOKING PREDICTION
≤ 1.0%	Fully Equalized	Balanced from first light. Optimal experience.
1.0 – 1.5%	Acceptable	Minor flavour timing shift. Small impact.
1.5 – 2.0%	Transitional	Flavour timing shift likely. Experience will change across the smoke.
> 2.0%	Unequalized	Imbalance likely. Correct before smoking.
 Delta is one of the most predictive indicators of cigar readiness and balance.		

Required competencies

- Measure cap and foot RH correctly.
- Calculate and interpret delta magnitude.
- Use direction to predict flavour evolution.
- Recommend corrective actions based on evidence.

Do not ignore direction, core behaviour or equilibration speed.

Core Professional Principles

Statements candidates should be able to defend



Five principles

- Cigars equalize axially rather than uniformly.
- The cap is a useful proxy for internal core moisture.
- Cap-foot delta predicts flavour timing.
- Direction matters as much as magnitude.
- Balanced cigars usually announce themselves early in the smoke.

Measure. Interpret. Act. Verify.

The Final Principle of Module 5

From subjective surprise to predictive service



Module 5 establishes

- Moisture gradients explain many inconsistent smoking experiences.
- Cap-foot diagnostics allow pre-light prediction.
- Flavour timing can be managed intentionally.
- Professional cigar advice becomes more evidence-based and repeatable.

"This cigar will open late and I can explain why."